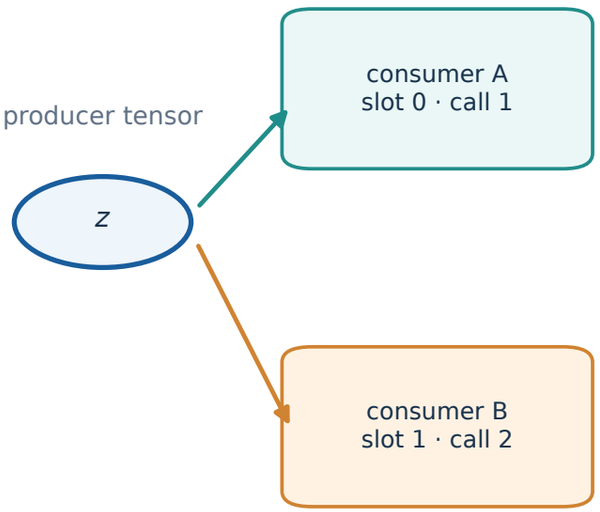
Valid differential cuts partition live paths exactly once

Consumer-use identity prevents shared producers, input slots and repeated invocations from being merged

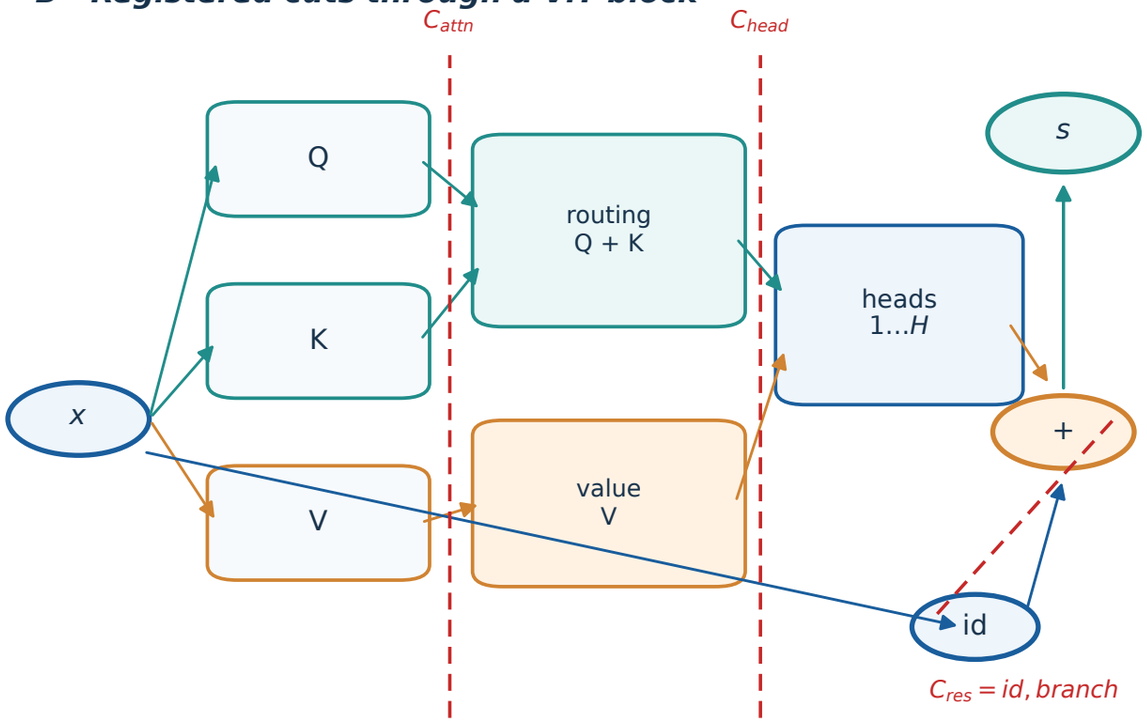
A · One producer, two concrete uses



$e = (\text{producer}, \text{consumer}, \text{input slot}, \text{invocation})$

same tensor ≠ same use edge

B · Registered cuts through a ViT block



$C_{attn} = \{Q, K, V\} = \{routing, value\}$

$C_{head} = \{h_1, \ldots, h_H\}$

C · Closure and audit

$$\mathcal{L}_{e,i} = x_i [J^{\top}_{z_e, x} \bar{z}^{(s)}_e]_i$$

$$\sum_{e \in C} \mathcal{L}_{e,i} = x_i \partial_i s$$

✓ each live path once

✗ omit nonzero edge

✗ duplicate identity

closed on raw signed input coordinates